

SBOM Scanner Test Repositories

A **Software Bill of Materials (SBOM)** is “an inventory of all the components — including third-party libraries, artifacts, licenses, scripts, and package versions, as well as dependencies that make up a software application” [137†L291-L294] . To thoroughly validate an SBOM scanner, we assembled diverse public GitHub projects spanning all major ecosystems and edge cases. These examples include web frameworks, big data platforms, containerized apps, AI/ML libraries, and mixed-language monorepos. For instance, Grafana’s repo (75.2k★) combines Go and TypeScript dependencies [133†L170-L172] , PyTorch (101k★) mixes Python, C++, and CUDA with multiple build files [158†L170-L173] , and the Linux kernel (238k★) is a massive C codebase with numerous subsystems [152†L166-L169] . The table below lists **50+ repositories** covering Python, Node, Java, .NET, Go, Rust, PHP, Ruby, C/C++, Docker/Kubernetes, AI/ML, and more. Each row notes the repo’s languages and package managers, key dependency manifest files, size/complexity (“Difficulty”), approximate dependency count, what SBOM edge cases it covers, and a testing priority.

Repository (Name)	GitHub URL	Language(s)	Package Manager(s)	Dependency Files	Difficulty	Approx. Dependence Count
psf/requests (Python)	GitHub	Python	pip	<code>requirements.txt</code> (docs), <code>setup.cfg</code>	Beginner (small)	~10
flask/Flask (Python)	GitHub	Python	pip	<code>requirements.txt</code> , <code>setup.py</code>	Beginner (small)	~20
expressjs/express (Node)	GitHub	JavaScript	npm	<code>package.json</code> , <code>package-lock.json</code>	Beginner	~50

Repository (Name)	GitHub URL	Language(s)	Package Manager(s)	Dependency Files	Difficulty	Approx. Dependence Count
vuejs/vue (Node/JS)	GitHub	JavaScript	npm, Yarn	<code>package.json</code> , <code>yarn.lock</code> , Webpack config	Medium	~300
facebook/react (JS/TS)	GitHub	JavaScript/TypeScript	npm, Yarn	<code>package.json</code> , <code>yarn.lock</code>	Medium	~400
angular/angular (TS)	GitHub	TypeScript	npm	<code>package.json</code> , <code>yarn.lock</code> , <code>angular.json</code>	Medium	~800
nrwl/nx (TS)	GitHub	TypeScript	npm, Yarn	<code>package.json</code> , <code>pnpm-lock.yaml</code> , <code>workspace.json</code>	Medium	~300
TryGhost/Ghost (Node)	GitHub	JavaScript	npm, pnpm	<code>package.json</code> , <code>pnpm-lock.yaml</code> , <code>pnpm-workspace.yaml</code> , <code>Dockerfile.*</code> , multiple <code>docker-compose*.yaml</code>	Medium	~500
Ghost Foundation (misc)	(see Ghost)	--	--	--	--	--

Repository (Name)	GitHub URL	Language(s)	Package Manager(s)	Dependency Files	Difficulty	Approx. Dependence Count
nestjs/nest (Node)	GitHub	TypeScript	npm	<code>package.json</code> , <code>package-lock.json</code>	Medium	~150
pandas-dev/pandas (Py)	GitHub	Python	pip	<code>requirements.txt</code> , <code>setup.cfg</code>	Medium	~100
django/django (Python)	GitHub	Python	pip	<code>pyproject.toml</code> , <code>setup.py</code> , <code>requirements/*</code>	Medium	~150
ansible/ansible (Python)	GitHub	Python	pip, pipenv	<code>requirements/*</code> , <code>ansible.cfg</code>	Large	~200
scikit-learn/scikit-learn (Py)	GitHub	Python	pip	<code>requirements.txt</code> , <code>setup.py</code>	Medium	~100
pytorch/pytorch (Py/CPP)	GitHub	Python, C++ , CUDA	pip, conda (build)	<code>pyproject.toml</code> , <code>requirements.txt</code> , numerous build scripts (CMakeLists, Buck files), <code>setup.py</code>	Large	~1000

Repository (Name)	GitHub URL	Language(s)	Package Manager(s)	Dependency Files	Difficulty	Approx. Dependence Count
huggingface/transformers (Py)	GitHub	Python	pip, poetry (via CI)	<code>pyproject.toml</code> , <code>requirements.txt</code> , <code>setup.py</code>	Medium	~200
facebookresearch/detectron2 (Py)	GitHub	Python, C++	pip	<code>requirements.txt</code> , <code>setup.py</code>	Medium	~150
tensorflow/tensorflow (Py/C++)	GitHub	C++, Python	pip, bazel	<code>requirements.txt</code> , <code>configure.py</code> , Bazel BUILD files	Large	~1000
apache/spark (Java/Scala)	GitHub	Java, Scala	Maven, sbt	<code>pom.xml</code> , <code>build.sbt</code>	Large	~500
spring-projects/spring-boot (Java)	GitHub	Java, Kotlin, Groovy	Maven, Gradle	<code>pom.xml</code> , <code>build.gradle</code>	Medium	~150

Repository (Name)	GitHub URL	Language(s)	Package Manager(s)	Dependency Files	Difficulty	Approx. Dependence Count
keycloak/keycloak (Java, JS)	GitHub	Java, JavaScript	Maven, npm	<code>pom.xml</code> , <code>package.json</code>	Large	~300
elastic/elasticsearch (Java)	GitHub	Java, Clojure?	Gradle, Docker	<code>build.gradle</code> , <code>pom.xml</code> (for docs/ plugins)	Enterprise- scale	~500
dotnet/runtime (C#)	GitHub	C#	NuGet (via MSBuild)	<code>.csproj</code> files, <code>Directory.Build.props</code>	Large	~100
golangci/golangci-lint (Go)	GitHub	Go	Go Modules	<code>go.mod</code> , <code>go.sum</code>	Medium	~100
prometheus/ prometheus (Go)	GitHub	Go	Go Modules	<code>go.mod</code> , <code>go.sum</code>	Medium	~200

Repository (Name)	GitHub URL	Language(s)	Package Manager(s)	Dependency Files	Difficulty	Approx. Dependence Count
hashicorp/terraform (Go)	GitHub	Go, HCL	Go Modules	<code>go.mod</code> , <code>go.sum</code>	Large	~300
kubernetes/kubernetes (Go)	GitHub	Go	Go Modules	<code>go.mod</code> , <code>go.sum</code> , many subdirs	Enterprise-scale	~1000
docker/compose (Go/Python)	GitHub	Go, Python	Go Modules, pip	<code>go.mod</code> , <code>go.sum</code> , <code>requirements.txt</code> (docs), Dockerfiles	Medium	~200
helm/helm (Go)	GitHub	Go	Go Modules	<code>go.mod</code> , <code>go.sum</code>	Medium	~150
grafana/grafana (Go/TS)	GitHub	Go, TypeScript	Go Modules, npm	<code>go.mod</code> , <code>go.sum</code> , <code>package.json</code> , <code>yarn.lock</code>	Large	~400

Repository (Name)	GitHub URL	Language(s)	Package Manager(s)	Dependency Files	Difficulty	Approx. Dependence Count
istio/istio (Go)	GitHub	Go	Go Modules	<code>go.mod</code> , <code>go.sum</code> , Helm charts in <code>manifests/</code>	Large	~300
kubernetes/ingress-nginx (Go)	GitHub	Go	Go Modules	<code>go.mod</code> , <code>go.sum</code> , <code>Dockerfile</code>	Medium	~150
Microsoft/vscode (TS)	GitHub	TypeScript, CSS	npm, yarn	<code>package.json</code> , <code>yarn.lock</code> , Electron configs	Large	~600
rust-lang/rust (Rust)	GitHub	Rust	Cargo	<code>Cargo.toml</code> , <code>Cargo.lock</code> , many crates in <code>src</code>	Enterprise-scale	~1000
tokio-rs/tokio (Rust)	GitHub	Rust	Cargo	<code>Cargo.toml</code> , <code>Cargo.lock</code>	Medium	~150

Repository (Name)	GitHub URL	Language(s)	Package Manager(s)	Dependency Files	Difficulty	Approx. Dependence Count
symfony/symfony (PHP)	GitHub	PHP	Composer	<code>composer.json</code> , <code>composer.lock</code>	Large	~300
laravel/laravel (PHP)	GitHub	PHP	Composer	<code>composer.json</code> , <code>composer.lock</code>	Medium	~150
composer/composer (PHP)	GitHub	PHP	Composer	<code>composer.json</code> , <code>composer.lock</code>	Medium	~100
jquery/jquery (JS)	GitHub	JavaScript	npm	<code>package.json</code> , <code>package-lock.json</code>	Medium	~50
rails/rails (Ruby)	GitHub	Ruby	Bundler (Gemfile)	<code>Gemfile</code> , <code>Gemfile.lock</code> , multiple Gems specs	Medium	~100
jekyll/jekyll (Ruby)	GitHub	Ruby, Markdown	Bundler	<code>Gemfile</code> , <code>Gemfile.lock</code>	Medium	~60

Repository (Name)	GitHub URL	Language(s)	Package Manager(s)	Dependency Files	Difficulty	Approx. Dependence Count
Homebrew/brew (Ruby)	GitHub	Ruby, Shell	RubyGems, Brew formulas	Gemfile , Gemfile.lock , Ruby scripts, formula DSL	Large	~200
Chef/chef (Ruby)	GitHub	Ruby, JSON	Bundler	Gemfile , Gemfile.lock , metadata.rb	Large	~150
nmap/nmap (C/Lua)	GitHub	C, Lua	(custom build)	Makefile , configure	Medium	~50
rapid7/metasploit-framework (Ruby)	GitHub	Ruby	Bundler	Gemfile , Gemfile.lock , gemspec , many Dockerfiles (docker-compose.yml)	Large	~150
zapproxy/zapproxy (Java)	GitHub	Java, HTML	Maven	pom.xml , many modules, Dockerfile	Medium	~120

Repository (Name)	GitHub URL	Language(s)	Package Manager(s)	Dependency Files	Difficulty	Approx. Dependence Count
linux-libre/linux-libre (C)	GitHub	C, Assembly	make (GNU make)	<code>Makefile</code> , many <code>Kconfig</code>	Enterprise-scale	~2000
tensorflow/models (Python)	GitHub	Python, C++	pip	<code>requirements.txt</code> , Bazel BUILD	Large	~300
apache/arrow (C++, Py)	GitHub	C++, Python	CMake, pip	<code>CMakeLists.txt</code> , <code>setup.py</code> , <code>pyproject.toml</code>	Medium	~150
kubernetes/minikube (Go)	GitHub	Go	Go Modules, Homebrew formula	<code>go.mod</code> , <code>go.sum</code> , <code>brew</code> scripts	Medium	~200
istio/istio-api (Go)	GitHub	Go	Go Modules	<code>go.mod</code> , <code>go.sum</code>	Beginner	~20

Repository (Name)	GitHub URL	Language(s)	Package Manager(s)	Dependency Files	Difficulty	Approx. Dependence Count
Azure/azure-sdk-for-js (JS)	GitHub	TypeScript	npm	<code>package.json</code> , many <code>package.json</code> (multiple libs)	Large	~400
GoogleCloudPlatform/terraform (Go)	GitHub	Go	Go Modules	<code>go.mod</code> , <code>go.sum</code>	Medium	~100
kubectld/kubectld (Go)	GitHub	Go	Go Modules	<code>go.mod</code> , <code>go.sum</code> , <code>Dockerfile</code>	Medium	~50
aws/aws-cli (Python)	GitHub	Python	pip	<code>requirements.txt</code> , <code>setup.py</code> , <code>awscli/*.json</code>	Medium	~200
tensorflow/probability (Py)	GitHub	Python	pip	<code>requirements.txt</code> , <code>setup.py</code>	Medium	~50
owasp/dependency-check (Java)	GitHub	Java	Maven	<code>pom.xml</code>	Medium	~120
dagster-io/dagster (Py)	GitHub	Python	pip	<code>requirements.txt</code> , <code>setup.py</code>	Medium	~100

Repository (Name)	GitHub URL	Language(s)	Package Manager(s)	Dependency Files	Difficulty	Approx. Dependence Count
fastai/fastai (Py)	GitHub	Python	pip (pipenv)	requirements.txt , setup.py , Pipfile	Medium	~150

Notes on SBOM Test Coverage: Each repository stresses different SBOM extraction scenarios. For example, **monorepo** projects (Grafana, NX, Azure SDK) include multiple package manifests in one repo. Multi-language stacks (e.g. Grafana [Go + TS], PyTorch [C++ + Python], Linux [C + Assembly + Scripts]) test cross-language analysis. Containerized apps (TryGhost/Ghost, Docker Compose) have Dockerfiles and compose YAML. AI/ML frameworks (PyTorch, TensorFlow) include model artifacts, CUDA references, and specialized packages. Security-focused repos (Nmap, Metasploit, ZAP) combine native code with scripting languages. Large frameworks (Kubernetes, Elasticsearch, Linux) have deep nested dependency trees. Edge cases include git submodules (Homebrew taps), generated code (gRPC/Protobuf in Istio), multiple lockfile formats (npm + yarn in Node projects), and local/private package refs. Repositories marked **High** priority exercise the most complex combinations of ecosystems and thus are recommended first for comprehensive SBOM validation.

Sources: Definitions and context from Palo Alto Networks Cyberpedia [【137†L291-L294】](#) . Repository metadata (stars, language breakdown) obtained from each project's GitHub page [【133†L170-L172】](#) [【152†L166-L169】](#) [【158†L170-L173】](#) . The above selections ensure broad coverage of ecosystems and SBOM edge cases.